

Lighting-Based Assembly Guidance for Reconfigurable Controller

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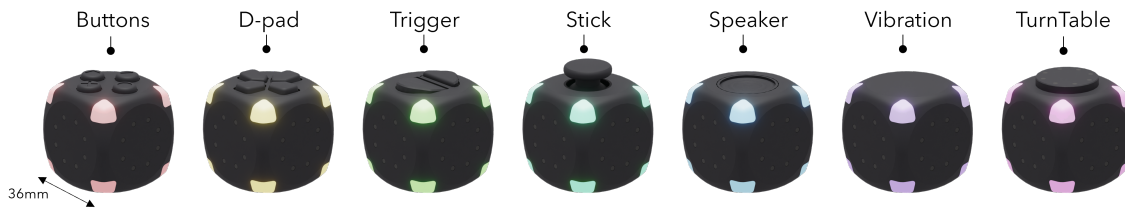


Figure 1: Image of developed devices and their functions of each module.

Abstract

This paper proposed a method for guiding users in reconfiguring a controller using multiple lighting parts on each functional module. Despite the high adaptability of these reconfigurable controllers, they require assembling skill for users. Our method uses lighting-based guidance by LEDs to allow the users to easily assemble a desired shape. In our method, lighting patterns specify not only the surface to be connected but also the direction to connect. This allows users to assemble modules that have input surfaces such as buttons and sticks without hesitation. To test the feasibility of incorporating this method into a game, we created a game in which reconfiguration occurs during play.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction devices**;

Keywords

Tangible interface, Reconfigurability, Lighting guidance

ACM Reference Format:

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1 Introduction

The rapid expansion of digital content and technological advancements has significantly diversified user bases and application needs. Consequently, there has been extensive research into developing reconfigurable controllers that users can customize by altering their shapes, unlike traditional controllers with fixed designs [Kitamura et al., 2000]. This innovation aims to address the increasingly diverse requirements of video game controllers.

Despite the high adaptability of these reconfigurable controllers, they present a challenge: users must assemble the controllers themselves. This task can be daunting for individuals with varying skill levels, particularly young children or those unfamiliar with assembling modular components, even when instructions are provided on a screen.

To address this issue, this paper proposes a method for guiding users in assembling reconfigurable controllers using multiple lighting parts. The reconfigurable controller in this paper comprises multiple modules, each equipped with input units for game control, such as buttons or sticks, and multiple LEDs as lighting parts. The guidance system operates by controlling these LEDs to direct

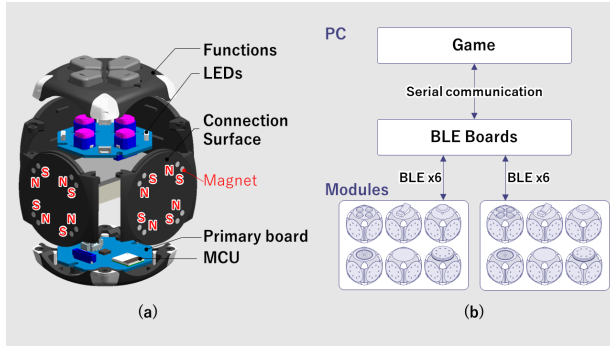


Figure 2: Device specifications.

the user through the assembly process, ensuring the controller is configured correctly and according to their preferences.

A relevant prior study is *VoxelHap* [Feick et al., 2023], where each surface of a Voxel features an LED, and the application indicates which surfaces should be connected. Our method, however, takes into account the orientation of the input components during gameplay. Thus, it not only specifies the connection surfaces but also the connection directions. This enhancement facilitates easier and more intuitive assembly of the controller.

2 Device specification

The reconfigurable controller module is designed as a cube with 36 mm sides and features seven types of input functions, as depicted in Fig. 1. Each vertex of the cube contains a full-color LED, amounting to eight LEDs per module. These LEDs are utilized to provide user guidance.

Fig. 2(a) illustrates a module with a D-pad as an input section. Eight magnets are positioned on each side of the module’s exterior (“connection surface”), excluding the input surfaces. The magnets are arranged with alternating north and south poles, enabling the connection surfaces to rotate 90 degrees during assembly. Internally, the module comprises two boards and a battery. The primary board includes an MCU with wireless capabilities, a push button for physical input, and eight full-color LEDs at each vertex, facilitating LED-based guidance. Additionally, Fig. 2(b) demonstrates the module’s connection system within a game application. All modules can be wirelessly controlled by the game application, allowing for guidance tailored to gameplay.

3 Guidance pattern

Examples of LED guidance patterns using the reconfigurable controller modules are provided. Fig. 3(a) demonstrates the specification of connection surfaces using LEDs. Fig. 3(b) expands on this by indicating both the connection surfaces and direction through changes in LED color. In Fig. 3(c), two LEDs are lighted in addition to Fig. 3(a) that gives the user a choice of surface to connect. Fig. 3(d) shows which module should be removed next. These LED lighting patterns are part of a set of guidance strategies that can be combined to facilitate the easy assembly of the controller as instructed.

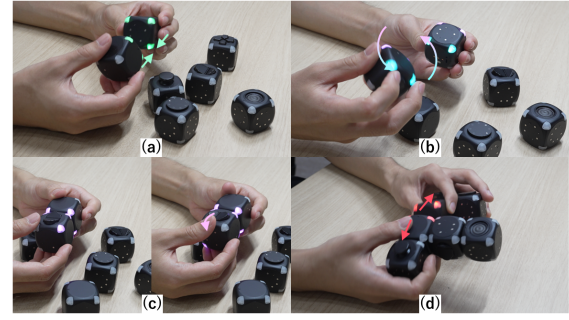


Figure 3: LED guidance patterns

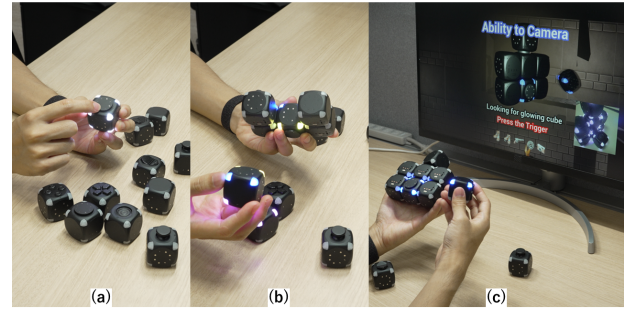


Figure 4: LED guidance patterns

4 Application

The effectiveness of this method was validated through the implementation of a game application utilizing lighting-based guidance. This game application employs multiple modules, requiring the user to progress by assembling to expand functionality of the controller. Only one module is available for game play when the game application starts as showed in 4(a). As in-game items are acquired, the number of available modules increases, necessitating appropriate assembly to advance. 4(b) presents that the proposed method, combined with on-screen instructions, guided users through the assembly process. Ultimately, the game involves assembling eight modules into a complex camera shape as depicted in 4(c). In our test play, participants successfully assembled the camera shape using the provided guidance patterns.

5 Conclusion & Future work

This paper proposed a method for guiding users in reconfiguring a controller using multiple light parts on a reconfigurable controller. The light parts provided guidance on connection surfaces and directions, enabling users to easily reconfigure the controller into the desired shape. Future work could involve recognizing the connection status between modules (e.g., connected or not connected, orientation, module ID) to offer timely and optimized guidance based on the user’s reconfiguration circumstances.

References

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